



How SOTB-Based MCUs Are Being Used To Harvest Energy

Silicon-on-thin-buried-oxide (SOTB) is a recent technology that helps obtain innovative energy-harvesting embedded controllers. These controllers can eliminate the need to use or replace batteries in the Internet of Things (IoT) devices



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A silicon-on-thin-buried-oxide (SOTB)-based embedded controller that combines silicon process and energy-harvesting knowledge has made appreciable technological progress in recent years. It totally eliminates the need for batteries in some products through harvesting ambient energy sources. Although sensors and wearable systems are expanding, giving more importance to the Internet of Things (IoT), effective battery management for such systems has so far remained challenging.

Some advantages of SOTB technology are:

- Provides extreme reduction in both active and standby current consumption

- Helps in energy harvesting by making use of such energy sources as light, vibration and flow

- Results in maintenance-free connected IoT devices

- Consumes power less than one-tenth of conventional low-power MCUs

- High speed with low leakage
- High operating frequency
- Optimal power control

Energy harvesting

Energy harvesting is also known as power harvesting, energy scavenging or ambient power. It is a process by which energy is derived from external sources, including solar, wind, heat, etc, and is captured and stored for small, wireless autonomous devices like wearable electronics and wireless sensor networks.

Present energy-conversion devices are inefficient. Motors get hot, so do power transistors and electric bulbs. In such cases, energy is wasted as heat. Energy-harvesting devices capture some of this wasted energy, convert it into electricity and put it to work.

Battery replacement may be difficult, expensive or even impossible in consumer and medical electronic devices. In such situations, energy harvesting comes to the rescue.

To solve power-related problems, energy harvesting has become an alternative and effective method. Harvesting ambient energy to power electronic devices has captured the imagination of electronics engineers for the longest time. However, it is only in recent years that this technology has



(Credit: www.businesswire.com)